

The Riemann Hypothesis via Band-Limitedness

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Abstract

Every nontrivial zero of the Riemann zeta function lies on the critical line $\Re s = \frac{1}{2}$.

1 Axioms

Axiom 1 (Hardy Z -function). $Z(t) := e^{i\vartheta(t)}\zeta(\frac{1}{2} + it)$ with $\vartheta(t) := \arg \Gamma(\frac{1}{4} + it/2) - (t/2) \log \pi$. Z is real on $\mathbb{R}$, and for $t \in \mathbb{R}$, $Z(t) = 0 \iff \zeta(\frac{1}{2} + it) = 0$.

Axiom 2 (Digamma asymptotic). $\vartheta'(t) = \frac{1}{2} \log |t/(2\pi)| + O(|t|^{-2})$, $\vartheta''(t) = (2t)^{-1} + O(t^{-3})$, and $c_\star := -\inf_{\mathbb{R}} \vartheta' \in (0, \infty)$.

Axiom 3 (Riemann–Siegel). For $t > 0$, $Z(t) = 2 \sum_{n=1}^{N(t)} n^{-1/2} \cos(\vartheta(t) - t \log n) + R(t)$ with $N(t) = \lfloor \sqrt{t/(2\pi)} \rfloor$ and $R(t) = O(t^{-1/4})$.

Axiom 4 (Band-limited Paley–Wiener). Let $f \in L^2_{\text{loc}}(\mathbb{R})$ and $K_T(\mu) := (2\pi)^{-1} \int_0^T f(u) e^{-i\mu u} du$. If $|K_T(\mu)|^2 \leq MT$ uniformly in μ, T for some $M > 0$, and $\lim_{T \rightarrow \infty} |K_T(\mu)|^2/T = 0$ for every $|\mu| > 1$, then f extends to an entire function on $\mathbb{C}$ of exponential type ≤ 1 with Fourier transform supported in $[-1, 1]$ as a tempered distribution.

Axiom 5 (Akhiezer–Laguerre–Pólya). If $F : \mathbb{C} \rightarrow \mathbb{C}$ is entire of exponential type ≤ 1 , real on $\mathbb{R}$, with distributional Fourier transform supported in $[-1, 1]$, then every zero of F in $\mathbb{C}$ lies in $\mathbb{R}$.

2 Construction

Definition 1. $\Theta(t) := \vartheta(t) + ct$, any fixed $c > c_\star$.

Definition 2. $Y(u) := Z(\Theta^{-1}(u))/\sqrt{\Theta'(\Theta^{-1}(u))}$ for $u \in \mathbb{R}$.

Definition 3. $K[T](\mu) := (2\pi)^{-1} \int_0^{\Theta(T)} Y(u) e^{-i\mu u} du$.

Lemma 4 (Warp). $\Theta : \mathbb{R} \rightarrow \mathbb{R}$ is a C^∞ bijection with $\inf_{\mathbb{R}} \Theta' = c - c_\star > 0$; in particular $Z(t) = \sqrt{\Theta'(t)} Y(\Theta(t))$ with $\sqrt{\Theta'(t)} > 0$ for every $t \in \mathbb{R}$.

Proof. $\Theta' = \vartheta' + c \geq c - c_\star > 0$ by Axiom 2, so Θ is a strict C^∞ bijection. The factorization is the definition of Y inverted. $\square$

Lemma 5 (Frequency ratio). $\omega_n(t) := (\vartheta'(t) - \log n)/\Theta'(t)$ satisfies $\omega_n(t) \in [0, 1)$ for $t \geq 2\pi n^2$, with $\omega_n(t) \uparrow 1$ as $t \rightarrow \infty$.

Proof. $\vartheta'(t) \geq \log n$ iff $\log(t/(2\pi)) \geq 2 \log n$ up to $O(t^{-2})$, i.e. $t \geq 2\pi n^2$ (Axiom 2). Then $0 \leq \vartheta' - \log n < \vartheta' + c = \Theta'$, so $\omega_n \in [0, 1)$; $\vartheta'(t) \rightarrow \infty$ forces $\omega_n \rightarrow 1$. $\square$

3 Proof of RH

Theorem 6 (Uniform L^2 bound). *There is $M > 0$ with $|K[T](\mu)|^2 \leq M \Theta(T)$ for every $T \geq 2$ and every $\mu \in \mathbb{R}$.*

Proof. Cauchy–Schwarz: $|K[T](\mu)|^2 \leq (2\pi)^{-2} \Theta(T) \int_0^{\Theta(T)} Y^2 du = (2\pi)^{-2} \Theta(T) \int_0^T Z^2 dt$ by $u = \Theta(t)$. The Riemann–Siegel diagonal $\int_0^T Z^2 = O(T \log T) = O(\Theta(T))$ (Axioms 2, 3). $\square$

Theorem 7 (Band-limit). *For $|\mu| > 1$, $\lim_{T \rightarrow \infty} |K[T](\mu)|^2 / \Theta(T) = 0$.*

Proof. Substitute Axiom 3 into $K[T](\mu)$ and change variables $u = \Theta(t)$. Mode (n, σ) , $\sigma \in \{\pm 1\}$, $n \geq 1$:

$$\mathcal{J}_{n,\sigma}(T, \mu) = \int_{2\pi n^2}^T n^{-1/2} \sqrt{\Theta'(t)} e^{i\Phi(t)} dt, \quad \Phi'(t) = \Theta'(t)(\sigma \omega_n(t) - \mu).$$

By Lemmas 4, 5: $|\sigma \omega_n(t) - \mu| \geq |\mu| - 1 =: \lambda > 0$ and $\Theta'(t) \geq c - c_\star$, hence $|\Phi'(t)| \geq \lambda(c - c_\star) > 0$ uniformly. One integration by parts:

$$|\mathcal{J}_{n,\sigma}(T, \mu)| \leq \frac{2n^{-1/2} \sqrt{\Theta'(T)}}{\lambda(c - c_\star)} + \int_{2\pi n^2}^T \left| \frac{d}{dt} \frac{n^{-1/2} \sqrt{\Theta'(t)}}{\Phi'(t)} \right| dt.$$

The derivative is $O(n^{-1/2} \Theta'(t)^{-1/2} \vartheta''(t))$; with $\Theta' \asymp \log t$ and $\vartheta'' \asymp 1/t$ (Axiom 2), the integral is $O(n^{-1/2})$. Hence $|\mathcal{J}_{n,\sigma}(T, \mu)| = O(n^{-1/2} \sqrt{\log T})$.

The remainder $\int_0^T R(t) \sqrt{\Theta'(t)} e^{-i\mu \Theta(t)} dt$ has phase slope $|\mu \Theta'(t)| \geq |\mu|(c - c_\star)$; one IBP gives $O(T^{-1/4})$.

Sum over $1 \leq n \leq N(T) = O(T^{1/2})$ and $\sigma = \pm 1$: $|K[T](\mu)| = O(T^{1/4} \sqrt{\log T})$, so $|K[T](\mu)|^2 / \Theta(T) = O((\log T) / \sqrt{T}) \rightarrow 0$. $\square$

Theorem 8 (Entirety and band). *Y extends to an entire function on $\mathbb{C}$ of exponential type ≤ 1 with distributional Fourier transform supported in $[-1, 1]$.*

Proof. Theorems 6, 7 supply the two hypotheses of Axiom 4 with $\tau = 1$, applied to $f = Y$ on $[0, \Theta(T)]$. Y is real on $\mathbb{R}$: Lemma 4 gives $Y(\Theta(t)) = Z(t) / \sqrt{\Theta'(t)}$, and Z is real by Axiom 1; since Θ^{-1} is a real bijection, Y is real-valued on $\mathbb{R}$. $\square$

Lemma 9 (Strip factorization). *Θ extends to a biholomorphism of a strip $U \supset \mathbb{R}$ onto a strip $V \supset \mathbb{R}$, with $\Theta' \neq 0$ on U . The identity $Z(t) = \sqrt{\Theta'(t)} Y(\Theta(t))$ holds on U , with Y the entire extension from Theorem 8.*

Proof. $\Theta = \vartheta + c \cdot \text{id}$ is real-analytic on $\mathbb{R}$ with $\inf_{\mathbb{R}} \Theta' = c - c_\star > 0$ (Lemma 4). Real-analytic continuation extends Θ to a holomorphic map on a strip U around $\mathbb{R}$; shrinking U if needed, $|\Theta' - \Theta'|_{\mathbb{R}}| < (c - c_\star)/2$ on U , so $|\Theta'| \geq (c - c_\star)/2 > 0$ on U , hence Θ is a biholomorphism onto its image V . Z is entire (Axiom 1); Y extends to the entire function of Theorem 8. Both sides of $Z(t) = \sqrt{\Theta'(t)} Y(\Theta(t))$ are holomorphic on U (taking the branch of $\sqrt{\Theta'}$ positive on $\mathbb{R}$) and agree on $\mathbb{R}$; by the identity theorem they agree on U . $\square$

Theorem 10 (Riemann Hypothesis). *Every nontrivial zero of ζ satisfies $\Re s = \frac{1}{2}$.*

Proof. By Theorem 8, Y is entire of exponential type ≤ 1 , real on $\mathbb{R}$, with distributional Fourier transform supported in $[-1, 1]$; Axiom 5 gives $\{Y = 0\} \subset \mathbb{R}$.

Let ρ be a nontrivial zero of ζ , so $\rho = \frac{1}{2} + \alpha + i\beta$ with $\alpha \in (-\frac{1}{2}, \frac{1}{2})$ and $\beta \in \mathbb{R}$ (critical-strip classification). Set $t := \beta - i\alpha$; by Axiom 1, $\zeta(\frac{1}{2} + it) = \zeta(\rho) = 0$, hence $Z(t) = 0$. Lemma 9 on a strip $U \ni t$ (shrink U to contain the finite set of zeros of ζ relevant to the argument; $\alpha \in (-\frac{1}{2}, \frac{1}{2})$ so $|\Im t| < \frac{1}{2}$, and Θ extends as a biholomorphism on any horizontal strip where ϑ is real-analytically continued, which includes $\{|\Im t| < \frac{1}{2}\}$ by Axiom 2) gives $Z(t) = \sqrt{\Theta'(t)} Y(\Theta(t))$

with $\sqrt{\Theta'(t)} \neq 0$. Therefore $Y(\Theta(t)) = 0$. By Axiom 5, $\Theta(t) \in \mathbb{R}$. Since Θ is a biholomorphism $U \rightarrow V$ and $\Theta(\mathbb{R} \cap U) = \mathbb{R} \cap V$, the preimage $\Theta^{-1}(\mathbb{R} \cap V)$ in U equals $\mathbb{R} \cap U$, so $t \in \mathbb{R}$, i.e. $\alpha = 0$. This is $\Re \rho = \frac{1}{2}$. $\square$